

My journey into AI/ML began with a curiosity about how intelligent systems make decisions and generate useful outputs from data. As a first-year B.Tech Information Technology student at Jaypee Institute of Information Technology, I started by learning Python, data structures, and the fundamentals of machine learning through Kaggle courses and hands-on projects.

One of my earliest AI projects was an AI Resume Roaster, where I built a web application that analyzes resumes and generates AI-powered feedback using large language models. Through this project, I learned prompt engineering, text processing, API integration, and how to design applications that combine traditional software engineering with modern AI capabilities.

More recently, I worked on HardwareAI, an AI-powered electronics recommendation system. For this project, I collected and processed product data, engineered features from product attributes, and trained a Random Forest model using Scikit-learn to generate recommendation scores. I also integrated Gemini AI to understand user requirements expressed in natural language and provide personalized recommendations. To gain experience with real-world deployment, I deployed the application on Render and built the supporting backend using Flask and MySQL. This project helped me understand the complete ML pipeline, from data preprocessing and model training to deployment and user interaction.

Alongside these projects, I have completed Kaggle's Introduction to Machine Learning and Intermediate Machine Learning courses, explored supervised learning algorithms, and practiced implementing machine learning workflows using Pandas, NumPy, and Scikit-learn.

However, I recognize that my understanding is still at an early stage. While I have built and deployed ML-powered applications, I want to deepen my knowledge of core machine learning concepts such as model evaluation, feature selection, optimization techniques, recommendation systems at scale, and modern deep learning approaches. I am also eager to learn how industry practitioners approach large-scale ML problems and how theoretical concepts translate into production systems.

I believe I am a strong fit for ML Summer School because I learn best by building. Whenever I encounter a new concept, I try to convert it into a working project rather than stopping at theory. This curiosity has pushed me to explore machine learning, software engineering, deployment, and AI applications beyond my coursework. I am excited about the opportunity to learn from experts, collaborate with like-minded students, and strengthen my foundation so I can build more impactful AI systems in the future.